

41 Revision history

Table 315. Document revision history

Date	Version	Changes
15-Sep-2011	1	Initial release.
19-Oct-2012	2	Updated reference documents and added Table 1: Applicable products on cover page. MEMORY: Updated Section 2: Memory and bus architecture. PWR: Updated VDDA and VREF+ decoupling capacitor in Figure 7: Power supply overview. VOSRDY bit changed to read-only in Section 5.4.3: PWR power control/status register (PWR_CSR). Removed VDDA in Section 5.2.3: Programmable voltage detector (PVD) and remove VDDA in PVDO bit description (Section 5.4.3: PWR power control/status register (PWR_CSR)). RCC: Updated Figure 20: Simplified diagram of the reset circuit and minimum reset pulse duration guaranteed by pulse generator restricted to internal reset sources. GPIOs: Updated Section 8.3.1: General-purpose I/O (GPIO). DMA: Updated direct mode description in Section 10.2: DMA main features. Updated direct mode description in Section : Memory-to-peripheral mode, and Section 10.3.12: FIFO/: Direct mode. Updated register access in Section 10.5: DMA registers. Modified Stream2 /Channel 2 in Table 42: DMA1 request mapping. Added note related to EN bit in Section 10.5.5: DMA stream x configuration register (DMA_SxCR) (x = 0..7). Updated definition of NDT[15:0] bits in Section 10.5.6: DMA stream x number of data register (DMA_SxNDTR) (x = 0..7). Interrupts: Updated number of maskable interrupts to 82 in Section 12.1.1: NVIC features. Updated Section 12.2: External interrupt/event controller (EXTI).

Table 315. Document revision history (continued)

Date	Version	Changes
19-Oct-2012	2 (continued)	ADC: Changed ADCCLK frequency to 30 MHz in Section 13.5: Channel-wise programmable sampling timee. Added recovery from ADC sequence in Section 13.8.1: Using the DMA and Section 13.8.2: Managing a sequence of conversions without using the DMA. Updated AWDIE in Section 13.13.2: ADC control register 1 (ADC_CR1). Added read and write access in Section 13.13: ADC registers. Advanced control timers (TIM1 and TIM8): Updated 16-bit prescaler range in Section 17.2: TIM1 and TIM8 main features. Updated OC1 block diagram in Figure 114: Output stage of capture/compare channel (channel 1 to 3). Updated update event generation in Upcounting mode and Downcounting mode in Section 17.3.2: Counter modes and Section 17.3.3: Repetition counter. Updated bits that control the dead-time generation in Section 17.3.11: Complementary outputs and dead-time insertion. Updated ways to generate a break in Section 17.3.12: Using the break function. Changed OCxREF to ETR in the example given in Section 17.3.13: Clearing the OCxREF signal on an external event and changed OCREF_CLR to ETRF in Figure 124: Clearing TIMx OCxREF. Updated configuration for example of counter operation in encoder interface mode in Section 17.3.16: Encoder interface mode. Added register access in Section 17.4: TIM1 and TIM8 registers. Changed definition of ARR[15:0] bits in Section 17.4.12: TIM1 and TIM8 auto-reload register (TIMx_ARR). Updated BKE definition in Section 17.4.18: TIM1 and TIM8 break and dead-time register (TIMx_BDTR).

Table 315. Document revision history (continued)

Date	Version	Changes
19-Oct-2012	2 (continued)	General purpose timers (TIM2 to TIM5): Removed all references to “repetition counter”. Added Figure 134: General-purpose timer block diagram. Updated 16-bit prescaler range in Section 18.2: TIM2 to TIM5 main features. External clock mode 2 ETR restricted to TIM2 to TIM4 in Section 18.3.3: Clock selection and Section 18.3.6: PWM input mode. Updated Section 18.3.9: PWM mode and Section 18.3.11: Clearing the OCxREF signal on an external event. Updated Figure 174: Master/Slave timer example to change ITR1 to ITR0. Updated read and write access to registers in Section 18.4: TIM2 to TIM5 registers. Restored bits 15 to 8 of TIMx_SMCR as well as Table 98: TIMx internal trigger connection in Section 14.4.3. Removed note 1 related to OC1M bits in Section 18.4.13: TIMx capture/compare register 1 (TIMx_CCR1). Updated TIMx_CCER bit description for TIM2 to TIM5 in Section 18.4.9: TIMx capture/compare enable register (TIMx_CCER). General purpose timers (TIM9 to TIM14): Updated 16-bit prescaler range in Section 19.2.1: TIM9/TIM12 main features and Section 19.2.2: TIM10/TIM11 and TIM13/TIM14 main features. Updated Figure 181: General-purpose timer block diagram (TIM10/11/13/14)) to remove TRGO trigger controller output. Added register access in Section 19.4: TIM9 and TIM12 registers and Section 19.5: TIM10/11/13/14 registers. Basic timers (TIM6 and TIM7): Removed all references to “repetition counter”. Updated 16-bit prescaler range in Section 20.2: TIM6 and TIM7 main features. HASH: Updated Section 25.3.1: Duration of the processing. RNG: Updated Section 24.1: RNG introduction.

Table 315. Document revision history (continued)

Date	Version	Changes
19-Oct-2012	2 (continued)	RTC: Updated Figure 237: RTC block diagram. Added formula to compute fck_apre in Figure 26.3.1: Clock and prescalers. Updated Section 26.3.9: RTC reference clock detection. Updated Section : RTC register write protection. Added RTC_SSR shadow register in Section 26.3.6: Reading the calendar. Updated description of DC[4:0] bits in Section 26.6.7: RTC calibration register (RTC_CALIBR). Renamed RTC_BKxR into RTC_BKPxR in Table 121: RTC register map and reset values. Added power-on reset value and changed reset value to system reset value in Section 26.6.11: RTC sub second register (RTC_SSR). Updated definition of ALARMOUTTYPE in Section 26.6.17: RTC tamper and alternate function configuration register (RTC_TAFCR). I2C: Modified Section 27.3.8: DMA requests. Updated bit 14 description in Section 27.6.3: I2C Own address register 1 (I2C_OAR1)). Updated definition of PE bit and note related to SWRST bit; moved note related to STOP bit to the whole register in Section 27.6.1: I2C Control register 1 (I2C_CR1). USART: Section 30.6.6: Control register 3 (USART_CR3): removed notes related to UART5 in DMAT and DMAR description. Updated TTable 142: Error calculation for programmed baud rates at fPCLK = 42 MHz or fPCLK = 84 Hz, oversampling by 16 and Table 143: Error calculation for programmed baud rates at fPCLK = 42 MHz or fPCLK = 84 MHz, oversampling by 8. SPI/I2S: Updated Section 28.1: SPI introduction. Changed I2S simplex communication/mode to half-duplex communication/mode. Updated flags in reception/transmission modes in Section 28.2.2: I2S features. Added Frame error flag in Table 128: I2S interrupt requests. Added register access in Section 28.5: SPI and I2S registers. Updated ERRIE definition in Section 28.5.2: SPI control register 2 (SPI_CR2). Renamed TIFRFE to FRE and definition updated in Section 28.5.3: SPI status register (SPI_SR).

Table 315. Document revision history (continued)

Date	Version	Changes
19-Oct-2012	2 (continued)	SDIO: Updated value and description for bits [45:40] and [7:1] in Table 176: R4 response. Updated value at bits [45:40] in Table 178: R5 response. CAN: Updated Figure 335: Dual CAN block diagram. Modified definition of CAN2SB bits in Section : CAN filter master register (CAN_FMR). Added register access in Section 32.9: CAN registers ETHERNET: Updated standard for precision networked clock synchronization in Section 33.1: Ethernet introduction and Section 33.2.1: MAC core features. Updated CR bit definition in Section : Ethernet MAC MII address register (ETH_MACMIIAR). Replace RTPR by PM bit in Table 192: Source address filtering. USB OTG FS Updated remote wake-up signaling bit and the resume interrupt in Section : Suspended state. Added peripheral register access in Section 34.16: OTG_FS control and status registers. Updated INEPTXSA description in OTG_FS_DIEPTXFx. Changed PHYSEL from bit 7 to bit 6 of the OTG_FS_GUSBCFG register. USB OTG HS Updated remote wake-up signaling bit and the resume interrupt in Section : Suspended state. Added peripheral register access in Section 35.12: OTG_HS control and status registers. Updated OTG_HS_CID reset value. Updated INEPTXSA description in OTG_HS_DIEPTXFx. Updated FSLSPCS for LS host mode, added PHYSEL in Section : OTG_HS host configuration register (OTG_HS_HCFG). Renamed PHYSEL into PHSEL and changed from bit 7 to bit 6 of the OTG_HS_GUSBCFG register. Updated OTG_HS_DIEPEACHMSK1 and OTG_HS_DOEPEACHMSK1 reset values.

Table 315. Document revision history (continued)

Date	Version	Changes
19-Oct-2012	2 (continued)	FSMC: Updated step b) in Section 36.3.1: Supported memories and transactions. Updated Table 196: FSMC_BTRx bit fields. Changed Clock divide ration min in Table 246: Programmable NAND/PC Card access parameters. Updated case of synchronous accesses in Section 36.5: NOR Flash/PSRAM controller. Changed minimum value for ADDSET to 0 in Table 203, Table 206, Table 207, Table 209, and Table 210. Move note from Figure 437: Mode1 write accesses and Figure 436: Mode1 read accesses. Move note from Figure 439: ModeA write accesses to Figure 438: ModeA read accesses. Updated Section : WAIT management in asynchronous accesses. Added register access in Section 36.5.6: NOR/PSRAM control registers and Section 36.6.2: NAND Flash / PC Card supported memories and transactions. Removed caution note in Section 36.6.1: External memory interface signalss. Updated Table 249: 16-bit PC Card. Updated step 3 in Section 36.6.4: NAND Flash operations. Updated Figure 455: Access to non 'CE don't care' NAND-Flash and note below in Section 36.6.5: NAND Flash prewait functionality. Updated access to I/O Space in Section 36.6.7: PC Card/CompactFlash operationss. Updated Table 251: 16-bit PC-Card signals and access type. Updated BUSTURN bit definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4)). Changed bits 16 to 19 to BUSTURN in Section : SRAM/NOR-Flash write timing registers 1..4 (FSMC_BWTR1..4) DEBUG: Updated Section 38.4.3: Internal pull-up and pull-down on JTAG pins. Electronic signature Updated Section 24: Device electronic signature introduction. Updated REV_ID[15:0] to add revision Z in Section 24.1: Unique device ID register (96 bits). Updated address and example in Section 24.2: Flash size.

Table 315. Document revision history (continued)

Date	Version	Changes
13-Nov-2012	3	Added STM32F42x and STM32F43x devices. Removed reference du Flash programming manual on cover page. Added Section 2.3.2: Flash memory overview and Section 3: Embedded Flash memory interface. Change RTC_50Hz into RTC_REFIN in Section 8.3.2: I/O pin multiplexer and mapping. Modified RTC alternate function naming in Section 8: General-purpose I/Os (GPIO) and Section 26: Real-time clock (RTC). Updated max. input frequency in Section 26.3.1: Clock and prescalers. Changed bit access type from 'rw' to 'w' and bit description updated in Section 10.5.3: DMA low interrupt flag clear register (DMA_LIFCR) and Section 10.5.4: DMA high interrupt flag clear register (DMA_HIFCR). Updated Figure 18: Frequency measurement with TIM5 in Input capture mode. Updated Section : Signals synchronization in Section 36: Flexible static memory controller (FSMC) Section 34: USB on-the-go full-speed (OTG_FS): updated Section Figure 389.: USB host-only connection, Section : VBUS valid, and Section : Host detection of a peripheral connection. Section 35: USB on-the-go high-speed (OTG_HS): updated Section : VBUS valid, and Section : Detection of peripheral connection by the host.
19-Feb-2013	4	Updated Section 2: Memory and bus architecture. Updated Figure 1: System architecture for STM32F405xx/07xx and STM32F415xx/17xx devices, and Figure 1: System architecture for STM32F405xx/07xx and STM32F415xx/17xx devices. Updated Table 4: Memory mapping vs. Boot mode/physical remap. Updated Figure 5: Sequential 32-bit instruction execution. removed note 1 from Table 13: Maximum program/erase parallelism. PWR: Updated Figure 7: Power supply overview. Updated Section 5.1.3: Voltage regulator. Added ADCDC1 bit in Section 5.5.1: PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx. SYSCFG: Added ADCxDC2 bit in Section 8.2.3: SYSCFG peripheral mode configuration register (SYSCFG_PMC) for STM32F42xxx and STM32F43xxx. ADC: Updated Section 13.9.3: Interleaved mode, Section 13.9.4: Alternate trigger mode, and Section 13.9.5: Combined regular/injected simultaneous mode to describe case of interrupted conversion. Updated Section : Temperature sensor, VREFINT and VBAT internal channels, Section 13.10: Temperature sensor, and Section 13.11: Battery charge monitoring. RTC: Updated BKP[31:0] bit description in Section 26.6.20: RTC backup registers (RTC_BKPxR). I2C: Updated Section 27.3.5: Programmable noise filter.

Table 315. Document revision history (continued)

Date	Version	Changes
19-Feb-2013	4 (continued)	FSMC: Updated write FIFO size in Section 36.1: FSMC main features. Updated Figure 434: FSMC block diagram. Updated Section 36.5.4: NOR Flash/PSRAM controller asynchronous transactions. Modified differences between Mode B and mode 1 in Section : Mode 2/B - NOR Flash. Modified differences between Mode C and mode 1 in Section : Mode C - NOR Flash - OE toggling. Modified differences between Mode D and mode 1 in Section : Mode D - asynchronous access with extended address. Updated NWAIT signal in Figure 449: Asynchronous wait during a read access, Figure 450: Asynchronous wait during a write access, Figure 451: Wait configurations, Figure 452: Synchronous multiplexed read mode - NOR, PSRAM (CRAM), and Figure 453: Synchronous multiplexed write mode - PSRAM (CRAM). Updated Table 195 to Table 214. Updated Section : SRAM/NOR-Flash chip-select control registers 1..4 (FSMC_BCR1..4). DEBUG Updated Figure 485: Block diagram of STM32 MCU and Cortex®-M4 with FPU-level debug support.

Table 315. Document revision history (continued)

Date	Version	Changes
15-Sep-2013	5	Added STM32F429xx and STM32F439xx part numbers. Replaced FSMC by FMC added Chrom-ART Accelerator, LCD-TFT and SAI interface. Updated Figure 2: System architecture for STM32F42xxx and STM32F43xxx devices. PWR: Updated Section 5.2.2: Brownout reset (BOR). Added note related to CSS enabling in Entering Stop mode sections in Section 5.3.4: Stop mode (STM32F405xx/07xx and STM32F415xx/17xx) and Section 5.3.5: Stop mode (STM32F42xxx and STM32F43xxx). Updated Stop mode entry in Table 27 and Table 29. Updated WUF bit definition in PWR_CSR registers. Changed CWUF and CSBF access type to 'w' in PWR_CR register. RCC: Updated LSEBYP bit definition in RCC_BDCR register. GPIOs: Updated description of OSPEEDR bits. Removed frequency value in description of OSPEEDR bits. Corrected typos: "IDRy[15:0]" replaced with "IDRy" in "GPIOx_IDR" register, "ODRy[15:0]" replaced with "ODRy" in "GPIOx_ODR" register and "OTy[1:0]" replaced with "OTy" in "GPIOx_OTYPER" register. DCMI: Updated Section 15.4: DCMI clocks. IWDG: Corrected Figure 213: Independent watchdog block diagram. RTC: Replaced all occurrences of "power-on reset" with "backup domain reset". Added caution note under Table 121: RTC register map and reset values. Changed SHPF bit type to 'r' in Section 26.6.4: RTC initialization and status register (RTC_ISR).. SPI: Updated definition of ERRIE bit in Section 28.5.2: SPI control register 2 (SPI_CR2). UART: Updated Section 30.3.8: LIN (local interconnection network) mode. Removed note in Section 30.3.13: Continuous communication using DMA. ETHERNET: Modified ETH_MACA0HR (and ETH_DMABMR reset values. Updated definitions of TSTS bit in ETH_MACSR, and TSTTR in ETH_PTPTSSR.

Table 315. Document revision history (continued)

Date	Version	Changes
15-Sep-2013	5 (continued)	USB OTG-FS: Removed note related to VDD range limitation below Figure 387: OTG A-B device connection and Figure 388: USB peripheral-only connection. FSMC: Updated Table 229, Table 232, Table 235, Table 239. Replaced all occurrences of DATALAT by DATLAT and SRAM/CRAM by SRAM/PSRAM in the whole section. Updated Section 36.1: FSMC main features. Changed bits 27 to 20 of FSMC_BWTR1..4 to reserved. Updated Section 36.6.7: PC Card/CompactFlash operations. Updated WREN bit in Table 231, Table 232, Table 233, Table 236, Table 239, Table 242, Table 245, and Table 249. Updated Section 36.5.4: NOR Flash/PSRAM controller asynchronous transactions, Section : SRAM/NOR-Flash chip-select control registers 1..4 (FSMC_BCR1..4), Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4) and Section : SRAM/NOR-Flash write timing registers 1..4 (FSMC_BWTR1..4). Updated definition of PWID in Section : PC Card/NAND Flash control registers 2..4 (FSMC_PCR2..4). FMC: Updated TRDC definition in Section : SDRAM Timing registers 1,2 (FMC_SDTR1,2). DEBUG: updated Figure 487: JTAG TAP connections.

Table 315. Document revision history (continued)

Date	Version	Changes
03-Feb-2014	6	Added note related to over-drive mode unavailable in 1.8 to 2.1 V VDD range in Section 3.5.1: Relation between CPU clock frequency and Flash memory read time. Updated maximum CPU frequency in Section 3.5.2: Adaptive real-time memory accelerator (ART Accelerator™). PWR: Updated Run mode/ over-drive mode in Section 5.1.4: Voltage regulator for STM32F42xxx and STM32F43xxx. RCC for STM32F42/43xx: Changed APB1/2 and AHB maximum frequencies.xw GPIOs: Updated Figure 27: Selecting an alternate function on STM32F42xxx and STM32F43xxx. DMA: Updated Section 10.3.7: Pointer incrementation and Section 10.3.11: Single and burst transfers.. INTERRUPTS AND EVENTS: Updated Table 62: Vector table for STM32F42xxx and STM32F43xxx. ADC: Updated Section 13.3.10: Discontinuous mode/Section : Regular group. DCMI: Updated Section 15.5.2: DCMI physical interface. LTDC: Updated resolution in note below Figure 82: LCD-TFT Synchronous timings. TIM1 and 8: Added note related to IC1F in Section 17.4.7: TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1). TIM2 to 5: Updated note related to IC1F in Section 18.4.7: TIMx capture/compare mode register 1 (TIMx_CCMR1).

Table 315. Document revision history (continued)

Date	Version	Changes
03-Feb-2014	6 (continued)	TIM9 to 14: Updated note related to IC1F in Section 19.5.5: TIM10/11/13/14 capture/compare mode register 1 (TIMx_CCMR1). RTC: Updated Section 26.3.11: RTC smooth digital calibration. Changed ALRBIE to ALRBE (bit 9) in Section 26.6.3: RTC control register (RTC_CR). I2C: Introduced Sm (standard mode) and Fm (fast mode) acronyms. FSMC: Updated BUSTURN definition in Table 245: FSMC_BTRx bit fields. FMC: Added Mobile LPDDR SDRAM. Updated Section : SDRAM initialization and Section : SDRAM controller read cycle and Figure 476: NAND Flash/PC Card controller waveforms for common memory access. Updated Section : SRAM/NOR-Flash chip-select control registers 1..4 (FMC_BCR1..4), Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4), Section : SRAM/NOR-Flash write timing registers 1..4 (FMC_BWTR1..4), Section : SDRAM Timing registers 1,2 (FMC_SDTR1,2) and Section : SDRAM Refresh Timer register (FMC_SDRTR). Removed mention “default valeur after reset” in Section : Common memory space timing register 2..4 (FMC_PMEM2..4), Section : Attribute memory space timing registers 2..4 (FMC_PATT2..4), and Section : I/O space timing register 4 (FMC_PIO4). Updated BUSTURN definition in Table 288: FMC_BTRx bit fields. Updated REV_ID bits in Section 38.6.1: MCU device ID code.

Table 315. Document revision history (continued)

Date	Version	Changes
15-May-2014	7	Embedded Flash memory interface: Updated Section : Physical remap in STM32F42xxx and STM32F43xxx. Updated bank 2 selection in Section 2.4: Boot configuration. Updated notes related to MERx and SER bits in Section : Mass Erase. Updated Section 3.7.5: Proprietary code readout protection (PCROP). Updated FLASH_OPTCR register reset value for STM32F42/43xx in Section 3.9.10: Flash option control register (FLASH_OPTCR) for STM32F42xxx and STM32F43xxx and Section 3.9.11: Flash option control register (FLASH_OPTCR1) for STM32F42xxx and STM32F43xxx. RCC (STM32F42/43xx): Updated PPLN caution note in Section 6.3.2: RCC PLL configuration register (RCC_PLLCFGR) SYSCFG Updated MEM_MODE in Section 9.3.1: SYSCFG memory remap register (SYSCFG_MEMRMP) LTDC: Changed resolution do XGA (1024x768) in Section 16.2: LTDC main features, Section 16.4.1: LTDC Global configuration parameters, and updated Section 16.7.3: LTDC Active Width Configuration Register (LTDC_AWCR). RTC Added note in Section 26.3.14: Calibration clock output. TIMER 1/8: Removed note related to IC1F bits in Section 17.4.7: TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1), TIM2 to 5: Replaced IC2S by CC2S. Updated Figure 161: Output stage of capture/compare channel (channel 1). Removed note related to IC1F bits in Section 18.4.7: TIMx capture/compare mode register 1 (TIMx_CCMR1). TIM9 to 14: Removed note related to IC1F bits in Section 19.5.5: TIM10/11/13/14 capture/compare mode register 1 (TIMx_CCMR1). USB OTG-HS: Updated DSPD definition in Section : OTG_HS device configuration register (OTG_HS_DCFG). FSMC Updated DATLAT bits definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4).

Table 315. Document revision history (continued)

Date	Version	Changes
15-May-2014	7 (continued)	FMC Updated Figure 474: Synchronous multiplexed read mode waveforms - NOR, PSRAM (CRAM). Updated DATLAT bits definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4). Updated FMC_BWTRx register address offsets in Table 297: FMC register map. DEBUG Added revision code '3' in Section : DBGMCU_IDCODE.

Table 315. Document revision history (continued)

Date	Version	Changes
14-Oct-2014	8	Memory and bus architecture: Updated Table 3: Memory mapping vs. Boot mode/physical remap in STM32F405xx/07xx and STM32F415xx/17xx and Table 4: Memory mapping vs. Boot mode/physical remap in STM32F42xxx and STM32F43xxx. RCC (STM32F40/41xx) and RCC (STM32F42/43xx): Removed all references to Flash programming manual. Changed RCC_AHB1LPENR, RCC_APB1LPENR, RCC_APB2LPENR, RCC_PLLI2SCFGR and RCC_APB2LPENR reset values. Updated access type to “r” for bits 24 to 31 in RCC_CSR. GPIOs: Updated Figure 27: Selecting an alternate function on STM32F42xxx and STM32F43xxx. IWDG Update note in Table 107: Min/max IWDG timeout period (in ms) at 32 kHz (LSI). CRYPTO and HASH Removed STM32F405/407xx and STM32F42xx from the whole sections. Removed STM32F405/407xx and STM32F42xx from the whole section. TIM10/11/13/14 Added TIMx_DIER description in Section 19.5: TIM10/11/13/14 registers. ETHERNET: Updated Table 187: Clock range. USB OTG FS: Removed TRDT formula in Section 34.17.7: Worst case response time and added Table 203: TRDT values. USB OTG HS: Removed TRDT formula in Section 35.13.8: Worst case response time and added Table 213: TRDT values. FSMC: Updated EXTMOD definition in Section : SRAM/NOR-Flash chip-select control registers 1..4 (FSMC_BCR1..4). Updated ADDSET definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4) and Section : SRAM/NOR-Flash write timing registers 1..4 (FSMC_BWTR1..4).

Table 315. Document revision history (continued)

Date	Version	Changes
14-Oct-2014	8 (continued)	FMC: Modified step 7 in Section : SDRAM initialization. Modified SDRAM refresh rate equations and example in Section : SDRAM Refresh Timer register (FMC_SDRTR) and updated definition of COUNT bits. Updated EXTMOD definition in Section : SRAM/NOR-Flash chip-select control registers 1..4 (FMC_BCR1..4). Updated ADDSET definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4) and Section : SRAM/NOR-Flash write timing registers 1..4 (FMC_BWTR1..4).

Table 315. Document revision history (continued)

Date	Version	Changes
16-Mar-2015	9	PWR: Updated Section 5.1.2: Battery backup domain. Updated Table 23: Low-power mode summary to add Return from ISR as entry condition. Added Section : Entering low-power mode and Section : Exiting low-power mode. Updated Section : Entering Sleep mode, Section : Exiting Sleep mode, Table 24: Sleep-now entry and exit and Table 25: Sleep-on-exit entry and exit. Updated Section : Entering Stop mode (for STM32F405xx/07xx and STM32F415xx/17xx), Section : Exiting Stop mode (for STM32F405xx/07xx and STM32F415xx/17xx) and Table 27: Stop mode entry and exit (for STM32F405xx/07xx and STM32F415xx/17xx). Updated Section : Entering Stop mode (STM32F42xxx and STM32F43xxx), Section : Exiting Stop mode (STM32F42xxx and STM32F43xxx) and Table 29: Stop mode entry and exit (STM32F42xxx and STM32F43xxx). Updated Section : Entering Standby mode, Section : Exiting Standby mode and Table 30: Standby mode entry and exit. RCC: Updated bits 24 to 31 access type in Section 7.3.21: RCC clock control & status register (RCC_CSR). GPIOs: Added port A reset value in Section 8.4.3: GPIO port output speed register (GPIOx_OSPEEDR) (x = A..I/J/K). DMA: Update FTH[1:0] description in Section 10.5.10: DMA stream x FIFO control register (DMA_SxFCR) (x = 0..7). TIM2/5: Register format changed to 32 bits instead of 16 in Section 18.4.10: TIMx counter (TIMx_CNT) and Section 18.4.12: TIMx auto-reload register (TIMx_ARR). TIM9 to 14: Updated Table 101: TIMx internal trigger connection WWDG: Updated Figure 214: Watchdog block diagram and Section 22.4: How to program the watchdog timeout. Updated Figure 215: Window watchdog timing diagram RNG: Replaced PLL48CLK by RNG_CLK in the whole section.

Table 315. Document revision history (continued)

Date	Version	Changes
16-Mar-2015	9 (continued)	I2C2: Updated <code>FREQ[5:0]</code> description in Section 27.6.2: I2C Control register 2 (<code>I2C_CR2</code>). USART: Removed note related to <code>RXNEIE</code> in Section : Reception using DMA FSMC: Updated Figure 474: Synchronous multiplexed read mode waveforms - NOR, PSRAM (CRAM). USB OTG FS Updated Table 203: TRDT values FMC Updated <code>FMC_NL</code> in Figure 456: FMC block diagram. Updated 'Memory wait' and 'Memory data bus high-z' parameters in Table 289: Programmable NAND Flash/PC Card access parameters. Updated Section : Common memory space timing register 2..4 (<code>FMC_PMEM2..4</code>). Updated Figure 476: NAND Flash/PC Card controller waveforms for common memory access. DEBUG: Updated <code>REV_ID[15:0]</code> and JTAG ID code in Section 38.6.1: MCU device ID code and Section 38.6.2: Boundary scan TAP, respectively

Table 315. Document revision history (continued)

Date	Version	Changes
28-Jul-2015	10	Embedded Flash memory interface Updated Section 3.7.5: Proprietary code readout protection (PCROP), Power controller (PWR) Added the last sentence in Subsection: Entering low-power mode of Section 5.3: Low-power modes, Added the bullet points about the interrupt in mode entry in Table 24: Sleep-now entry and exit, Table 25: Sleep-on-exit entry and exit, Table 27: Stop mode entry and exit (for STM32F405xx/07xx and STM32F415xx/17xx), Table 29: Stop mode entry and exit (STM32F42xxx and STM32F43xxx) Added the last point to Mode entry, on return from ISR in Table 30: Standby mode entry and exit, Added the note in Section: Entering sleep mode in Section 5.3.3: Sleep mode. General-purpose I/Os (GPIO) Updated OSPEED[1:0] definition of GPIOx_OSPEEDR register in Section 8.4.3: GPIO port output speed register (GPIOx_OSPEEDR) (x = A..I/J/K) LCD-TFT Controller (LTDC) Corrected the bit field for WHSTPOS in the second bullet point in Section: Window in Section 16.4.2: Layer programmable parameters. Advanced-control timers (TIM1&TIM8) Added the note in Section 17.3.20: Timer synchronization, Updated ETF[3:0] description in Section 17.4.3: TIM1 and TIM8 slave mode control register (TIMx_SMCR), Updated IC1F[3:0] description in Section 17.4.7: TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1), Added the note to MMS2 bit description in Section 17.4.8: TIM1 and TIM8 capture/compare mode register 2 (TIMx_CCMR2), Added the note to SMS[2:0] bit description in Section 17.4.3: TIM1 and TIM8 slave mode control register (TIMx_SMCR). General-purpose timers (TIM2 to TIM5) Added the note in Section 18.3.15: Timer synchronization, Updated SMS[2:0] description in Section 18.4.3: TIMx slave mode control register (TIMx_SMCR), Added the note to MMS2 bit description in Section 18.4.2: TIMx control register 2 (TIMx_CR2), Added the note to SMS[2:0] bit description in Section 18.4.3: TIMx slave mode control register (TIMx_SMCR).

Table 315. Document revision history (continued)

Date	Version	Changes
28-Jul-2015	10 (Continued)	General-purpose timers (TIM9 to TIM14) Added the note in Section 19.3.12: Timer synchronization (TIM9/12), Added the note to MMS2 bit description, Added the note to SMS[2:0] bit description in Section 19.4.2: TIM9/12 slave mode control register (TIMx_SMCR). Window watchdog (WWDG) Updated.Figure 214: Watchdog block diagram Controller area network (bxCAN) Replaced tCAN with tq, Flexible static memory controller (FSMC) Added the paragraph about Cross boundary page for Cellular RAM 1.5 in Section 36.5.5: Synchronous transactions, Updated MEMHIZx, MEMHOLDx, MEMSETx bit field descriptions for FSMC_PME2..4 register in Section 36.5.5: Synchronous transactions, Updated ATTSET, ATTHOLD, ATTHIZ bit field descriptions for FSMC_PATT2..4 register in Section 36.5.5: Synchronous transactions, Updated IRS and IFS bit descriptions for FMC_SR2..4 in Section 36.5.5: Synchronous transactions, Renamed ADDSET as ADDSET[3:0] and MTYP as MTYP[1:0], Addition of CPSIZE in FSMC_BCRx bit fields in Table 226: FSMC_BCRx bit fields, Table 228: FSMC_BCRx bit fields, Table 231: FSMC_BCRx bit fields, Table 234: FSMC_BCRx bit fields, Table 237: FSMC_BCRx bit fields, Table 240: FSMC_BCRx bit fields, Table 242: FSMC_BCRx bit fields, Added CPIZE[2:0] in FMC_BCR1...4 registers in ,Section 36.5.6: NOR/PSRAM control registers Section NOR/PSRAM control re Added CPSIZE[2:0] for FMC_BCRx registers in Section 36.6.9: FSMC register map.

Table 315. Document revision history (continued)

Date	Version	Changes
28-Jul-2015	10 (Continued)	Flexible memory controller (FMC) Added the paragraph about Cross boundary page for Cellular RAM 1.5 in Section 37.5.5: Synchronous transactions, Updated BUSTURN bit field description for FMC_BTR1..4 register in Section 37.5.6: NOR/PSRAM controller registers, Updated MEMHIZx, MEMHOLDx, MEMSETx bit field descriptions for FMC_PME2..4 register in Section 37.6.8: NAND Flash/PC Card controller registers, Updated ATTSET, ATTHOLD, ATTHIZ bit field descriptions for FMC_PATT2..4 register in Section 37.6.8: NAND Flash/PC Card controller registers, Updated IRS and IFS bit descriptions for FMC_SR2..4 in Section 37.6.8: NAND Flash/PC Card controller registers, Updated the section SDRAM initialization with the last item in the numbered list in Section 37.7.5: SDRAM controller registers, Renamed ADDSET as ADDSET[3:0] and MTYP as MTYP[1:0], Addition of CPSIZE in Table 269: FMC_BCRx bit fields, Table 271: FMC_BCRx bit fields, Table 274: FMC_BCRx bit fields, Table 277: FMC_BCRx bit fields, Table 280: FMC_BCRx bit fields, Table 283: FMC_BCRx bit fields, Table 285: FMC_BCRx bit fields, Table 287: FMC_BCRx bit fields, Added the paragraph about Cross boundary page for Cellular RAM 1.5 in Section 37.5.5: Synchronous transactions, Added CPIZE[2:0] in FMC_BCR1...4 registers in Section 37.5.6: NOR/PSRAM controller registers, Added CPSIZE[2:0] for FMC_BCRx registers in Section 37.8: FMC register map.

Table 315. Document revision history (continued)

Date	Version	Changes
20-Oct-2015	11	Reset and clock controller (RCC) Updated STM32F405/407/415/417xx Figure 21: Clock tree. Updated General purpose I/O (GPIOs) Changed definition of OSPEEDR bits in Section 8.4.3: GPIO port output speed register (GPIOx_OSPEEDR) (x = A..I/J/K). LCD-TFT display controller (LTDC): Changed LRDC_IER into LTDC_IER in Section 16.5: LTDC interrupts. Updated AHBP[11:0], AAV[11:0 and TOTALW[11:0 in Table 92: LTDC register map and reset values. Controller area network (bxCAN): Updated Section 32.3.4: Acceptance filters and Section 32.7.4: Identifier filtering. Flexible static memory controller (FSMC) Updated BUSTURN description in Section : SRAM/NOR-Flash write timing registers 1..4 (FSMC_BWTR1..4) and Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4) Updated note related to IRS and IFS bits in Section : FIFO status and interrupt register 2..4 (FSMC_SR2..4). Flexible memory controller (FMC) Updated paragraph related to the cacheable read FIFO in Section : SDRAM controller read cycle. Updated BUSTURN description in Section : SRAM/NOR-Flash write timing registers 1..4 (FMC_BWTR1..4) and Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4). Updated note related to IRS and IFS bits in Section : FIFO status and interrupt register 2..4 (FMC_SR2..4). Real-time clock (RTC2) Updated WUCKSEL prescaler input in Figure 237: RTC block diagram. Updated 3rd step in Section : Programming the wakeup timer. Updated WUTWF bit definition in Section 26.6.4: RTC initialization and status register (RTC_ISR).

Table 315. Document revision history (continued)

Date	Version	Changes
17-May-2016	12	Flash memory interface Removed note related to boot from Bank 2 in Section 2.4: Boot configuration. Updated notes in Section 3.7.3: Read protection (RDP). Changed number of LATENCY bits in Section 3.9.2: Flash access control register (FLASH_ACR) for STM32F42xxx and STM32F43xxx In Table 9: 1 Mbyte dual bank Flash memory organization (STM32F42xxx and STM32F43xxx): updated sector 19 size and option bytes (bank 2) address range. Power control (PWR) Removed reference to low-power mode in Section 5.1.4: Voltage regulator for STM32F42xxx and STM32F43xxx, Section : Entering Stop mode (STM32F42xxx and STM32F43xxx) and Section : Exiting Stop mode (STM32F42xxx and STM32F43xxx). Analog-to-digital converter (ADC) Added note related to ADC_HTR and ADC_LTR register programming in Section 13.13.7: ADC watchdog higher threshold register (ADC_HTR) and Section 13.13.8: ADC watchdog lower threshold register (ADC_LTR). Chrom-Art Accelerator™ controller (DMA2D) Updated Section 11.3.12: DMA2D transfer control (start, suspend, abort and completion). Section 11.5.8: DMA2D foreground PFC control register (DMA2D_FGPFCCR): updated START bit access type Section 11.5.10: DMA2D background PFC control register (DMA2D_BGPFCCR): updated START bit access and description. LCD-TFT controller (LTDC) Updated Section 16.3.2: LTDC reset and clocks. Modified LCD_DE description in Table 89: LCD-TFT pins and signal interface. Modified Section 16.7.15: LTDC Layerx Window Horizontal Position Configuration Register (LTDC_LxWHPCR) (where x=1..2) and Section 16.7.16: LTDC Layerx Window Vertical Position Configuration Register (LTDC_LxWVPCR) (where x=1..2). General-purpose timers (TIM2 to TIM5) Updated Section 18.4.11: TIMx prescaler (TIMx_PSC). General-purpose timers (TIM9 to TIM14) Added OPM bit in Section 19.5.1: TIM10/11/13/14 control register 1 (TIMx_CR1). Updated Section 19.4.9: TIM9/12 prescaler (TIMx_PSC) and Section 19.5.8: TIM10/11/13/14 prescaler (TIMx_PSC).

Table 315. Document revision history (continued)

Date	Version	Changes
17-May-2016	12 (continued)	General-purpose timers (TIM6 and TIM7) Updated Section 20.4.7: TIM6 and TIM7 prescaler (TIMx_PSC). Real-time clock (RTC) Updated conditions for running under System reset in Section 26.3.7: Resetting the RTC. Updated Section 26.3.14: Calibration clock output. Added note related to TSE in Section 26.6.3: RTC control register (RTC_CR). Updated caution note related to TAMP1TRG in Section 26.6.17: RTC tamper and alternate function configuration register (RTC_TAFCR) register. Universal synchronous asynchronous receiver transmitter (USART) Replaced all occurrences of nCTS by CTS, nRTS by RTS and SCLK by CK. Flexible static memory controller (FSMC) Updated Section 36.3: AHB interface. Added note related to the hold phase delay below Figure 454: NAND/PC Card controller timing for common memory access. Updated Section 36.6.5: NAND Flash prewait functionality. Updated BUSTURN description in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4). Updated MEMHOLDx in Section : Common memory space timing register 2..4 (FSMC_PMEM2..4) and ATTHOLD in Section : Attribute memory space timing registers 2..4 (FSMC_PATT2..4). Flexible memory controller (FMC) Updated Section 37.3: AHB interface. Added note related to the hold phase delay below Figure 476: NAND Flash/PC Card controller waveforms for common memory access. Updated Section 37.6.5: NAND Flash prewait functionality. Updated BUSTURN description in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4). Updated MEMHOLDx in Section : Common memory space timing register 2..4 (FMC_PMEM2..4) and ATTHOLD in Section : Attribute memory space timing registers 2..4 (FMC_PATT2..4). Debug (DBG) Updated value to be programmed to the ETM Trace Start/stop register to enable the trace in Section 38.15.4: ETM configuration example.

Table 315. Document revision history (continued)

Date	Version	Changes
20-Sep-2016	13	Analog-to-digital converter (ADC) Updated DMA mode 1 and DMA mode 3 description in Section 13.9: Multi ADC mode. LCD-TFT controller Updated values to be programmed to LTDC_SSCR in Section : Example of Synchronous timings configuration Updated Section 16.4.2: Layer programmable parameters/Windowing. Advanced-control timers (TIM1 and TIM8) Updated Section 17.3.21: Debug mode. Extended Section 17.4.20: TIM1 and TIM8 DMA address for full transfer (TIMx_DMAR) to 32 bits. Updated Table 95: Output control bits for complementary OCx and OCxN channels with break feature output state for MOE = 0. Updated TIM1 and TIM8 auto-reload register (TIMx_ARR) reset value. Updated TIMx_CCR1/2/3/4 description when CC1 channel is configured as inputs and changed bit access type to rw/ro. General-purpose timers (TIM2 to TIM5) Updated TIMx auto-reload register (TIMx_ARR) reset value. Updated TIMx_CCR1/2/3/4 description when CC1 channel is configured as inputs and changed bit access type to rw/ro. General-purpose timers (TIM9 to TIM14) Updated TIM9/12 auto-reload register (TIMx_ARR) and TIM10/11/13/14 auto-reload register (TIMx_ARR) reset value. Updated TIMx_CCR1 description when CC1 channel is configured as inputs and changed bit access type to rw/ro. Basic timers (TIM6 to TIM7) Updated TIM6 and TIM7 auto-reload register (TIMx_ARR). Secure digital input/output interface (SDIO) Updated Section 31.1: SDIO main features up to 50 MHz. Updated Section 31.3: SDIO functional description SDIO_CK description. Updated note removing 48 MHz in Section 31.9.1: SDIO power control register (SDIO_POWER), Section 31.9.2: SDI clock control register (SDIO_CLKCR), Section 31.9.4: SDIO command register (SDIO_CMD) and Section 31.9.9: SDIO data control register (SDIO_DCTRL).

Table 315. Document revision history (continued)

Date	Version	Changes
20-Sep-2016	13 (continued)	FMC Update BUSTURN bit description in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4) and Section : SRAM/NOR-Flash write timing registers 1..4 (FMC_BWTR1..4). Debug support Specified behavior of timers with complementary outputs in Section 38.16.2: Debug support for timers, watchdog, bxCAN and I2C. Updated DBG_TIMx_STOP bit description in Section 38.16.4: Debug MCU APB1 freeze register (DBGMCU_APB1_FZ) and Section 38.16.4: Debug MCU APB1 freeze register (DBGMCU_APB1_FZ). Electronic signature Updated Section 24.1: Unique device ID register (96 bits).
21-Apr-2017	14	Updated: – Section 5.5.2: PWR power control/status register (PWR_CSR) for STM32F42xxx and STM32F43xxx – Section 6.3.14: RCC APB2 peripheral clock enable register (RCC_APB2ENR) – Section 14.3.5: DAC output voltage – Section 38.6.1: MCU device ID code – Figure 237: RTC block diagram Deleted: – Section 7.3.15: RCC APB2 peripheral clock enable register(RCC_APB2ENR)
18-Jul-2017	15	Updated: – Section 3.9.10: Flash option control register (FLASH_OPTCR) for STM32F42xxx and STM32F43xxx – OTG_FS USB configuration register (OTG_FS_GUSBCFG) – Table 142: Error calculation for programmed baud rates at fPCLK = 42 MHz or fPCLK = 84 Hz, oversampling by 16 and Table 143: Error calculation for programmed baud rates at fPCLK = 42 MHz or fPCLK = 84 MHz, oversampling by 8.
23-Apr-2018	16	Updated: – Section 30.6.1: Status register (USART_SR) – Section 34.16.4: Device-mode registers – Section 34.17.6: Operational model – Section 35.12.4: Device-mode registers – Section 34: USB on-the-go full-speed (OTG_FS) – Table 199: Host-mode control and status registers (CSRs) – Table 205: OTG_FS register map and reset values – Table 210: Device-mode control and status registers – Table 215: OTG_HS register map and reset values Added: – Figure 412: SOF trigger output to TIM2 ITR1 connection – RXOLNY register changed from SPI_CR2 to SPI_CR1 in Section 28.3.4: Configuring the SPI for half-duplex communication and Unidirectional receive-only procedure (BIDIMODE=0 and RXONLY=1)

Table 315. Document revision history (continued)

Date	Version	Changes
07-Jun-2018	17	Updated: – Figure 16: Clock tree (STM32F42xxx an STM32F43xxx) and Figure 21: Clock tree (STM32F405xx/07xx and STM32F415xx/17xx) – Figure 27: Selecting an alternate function on STM32F42xxx and STM32F43xxx – Table 61: Vector table for STM32F405xx/07xx and STM32F415xx/17xx and Table 62: Vector table for STM32F42xxx and STM32F43xxx – Section 29.17.5: SAI xInterrupt mask register (SAI_xIM) where x is A or B – Section 38.6.1: MCU device ID code

Table 315. Document revision history (continued)

Date	Version	Changes
25-Feb-2019	18	Section 6: Reset and clock control for STM32F42xxx and STM32F43xxx (RCC) Updated OTGHSULPILPEN bit description in RCC AHB1 peripheral clock enable in low power mode register (RCC_AHB1LPENR) and OTGHSULPIEN bit description in RCC AHB1 peripheral clock register (RCC_AHB1ENR). Section 7: Reset and clock control for STM32F405xx/07xx and STM32F415xx/17xx(RCC): Updated RCC APB2 peripheral clock enabled in low power mode register (RCC_APB2LPENR) reset value. Updated OTGHSULPILPEN bit description in RCC AHB1 peripheral clock enable in low power mode register (RCC_AHB1LPENR) and OTGHSULPIEN bit description in RCC AHB1 peripheral clock enable register (RCC_AHB1ENR). Section 13: Analog-to-digital converter (ADC) Update Section : Dual ADC mode. Section 17: Advanced-control timers (TIM1 and TIM8) Updated Figure 113: Capture/compare channel 1 main circuit. Figure 19: General-purpose timers (TIM9 to TIM14) Updated Figure 194: Capture/compare channel 1 main circuit. Section 34: USB on-the-go full-speed (OTG_FS) Updated Section : SETUP and OUT data transfers and Updated Section : IN data transfers. Modified Table 200: Device-mode control and status registers. Section 35: USB on-the-go high-speed (OTG_HS) Updated Table 208: Core global control and status registers (CSRs) and Table 210: Device-mode control and status registers. Updated Section : OTG_HS device IN endpoint transmit FIFO size register (OTG_HS_DIEPTXFx) (x = 1..5, where x is the FIFO_number), Section : OTG device endpoint-x control register (OTG_HS_DIEPCTLx) (x = 0..5, where x = Endpoint_number), Section : OTG_HS device endpoint-x control register (OTG_HS_DOEPCTLx) (x = 1..5, where x = Endpoint_number), Section : OTG_HS device endpoint-x interrupt register (OTG_HS_DIEPINTx) (x = 0..5, where x = Endpoint_number), Section : OTG_HS device endpoint-x interrupt register (OTG_HS_DOEPINTx) (x = 0..5, where x = Endpoint_number), Section : OTG_HS device endpoint-x DMA address register (OTG_HS_DIEPDMAx / OTG_HS_DOEPDMAx) (x = 0..5, where x = Endpoint_number). Updated Section : SETUP and OUT data transfers and Section : IN data transfers. Section 38: Debug support (DBG) Updated REV_ID in DBGMCU_CR register.

Table 315. Document revision history (continued)

Date	Version	Changes
25-Feb-2021	19	Updated: – Section 2: Memory and bus architecture: – Figure 1: System architecture for STM32F405xx/07xx and STM32F415xx/17xx devices – Figure 2: System architecture for STM32F42xxx and STM32F43xxx devices – Section 3: Embedded Flash memory interface: – Table 16: Description of the option bytes (STM32F405xx/07xx and STM32F415xx/17xx) – Table 17: Description of the option bytes (STM32F42xxx and STM32F43xxx) – Section 12: Interrupts and events: – Table 61: Vector table for STM32F405xx/07xx and STM32F415xx/17xx – Table 62: Vector table for STM32F42xxx and STM32F43xxx – Section 13: Analog-to-digital converter (ADC): – Section 13.3.5: Continuous conversion mode – Section 13.8.1: Using the DMA – Section 13.10: Temperature sensor – Section 17: Advanced-control timers (TIM1 and TIM8): – Figure 86: Advanced-control timer block diagram – Section 17.4.9: TIM1 and TIM8 capture/compare enable register (TIMx_CCER) – Section 27: Inter-integrated circuit (I2C) interface: – Section 27.6.2: I2C Control register 2 (I2C_CR2) – Section 27.6.8: I2C Clock control register (I2C_CCR) – Section 31: Secure digital input/output interface (SDIO): – Section 31.9.2: SDI clock control register (SDIO_CLKCR) – Section 32: Controller area network (bxCAN): – Section 32.4.1: Initialization mode – Section 33: Ethernet (ETH): media access control (MAC) with DMA controller: – Section : TxDMA operation: default (non-OSF) mode – Section : Normal Tx DMA descriptors – Section 35: USB on-the-go high-speed (OTG_HS): – Section : Host port power – Section : OTG_HS core interrupt register (OTG_HS_GINTSTS) – Section : OTG_HS all endpoints interrupt mask register (OTG_HS_DAINTRMSK) – Table 215: OTG_HS register map and reset values – Section 35.13.2: Host initialization – Figure 416: Transmit FIFO write task – Figure 417: Receive FIFO read task – Figure 426: Receive FIFO packet read in slave mode – Figure 427: Processing a SETUP packet – Section 38: Debug support (DBG): – Section : DBGMCU_IDCODE

Table 315. Document revision history (continued)

Date	Version	Changes
05-Feb-2024	20	Section : Introduction: – Mentioned that the microcontrollers include ST state-of-the-art patented technology – Added errata sheets in the list of reference documents. Section 2: Memory and bus architecture – <i>Section 2.3.1: Embedded SRAM</i>, updated the SRAM that can be accessed through System or I-Code/D-Code bus. Section 3.4: Embedded flash memory in STM32F42xxx and STM32F43xxx – Updated Bank 2 section 14 base address in <i>Table 12: Flash module - 2 Mbyte dual bank organization (STM32F42xxx and STM32F43xxx)</i>. – Specified that dual bank organization is not available on 512 Kbyte devices, and added <i>Table 16: 512 Kbyte single bank flash memory organization (STM32F42xxx and STM32F43xxx)</i>. – Updated <i>Section 3.8.2: Program/erase parallelism</i>. – In <i>Section 3.9.3: Read protection (RDP)</i>, added note concerning RDP when debugger is connected through JTAG/SWD. Section 5: Power controller (PWR) – Updated <i>Figure 12: Power-on reset/power-down reset waveform</i>. – Updated no external battery use case in <i>Section 5.1.2: Battery backup domain</i>. – Updated DBP bit description in <i>PWR power control register (PWR_CR)</i> for <i>STM32F405xx/07xx</i> and <i>STM32F415xx/17xx</i> and <i>PWR power control register (PWR_CR)</i> for <i>STM32F42xxx</i> and <i>STM32F43xxx</i>. – Updated BRE bit description in <i>PWR power control/status register (PWR_CSR)</i> for <i>STM32F405xx/07xx</i> and <i>STM32F415xx/17xx</i> and <i>PWR power control/status register (PWR_CSR)</i> for <i>STM32F42xxx</i> and <i>STM32F43xxx</i>. Section 6: Reset and clock control for STM32F42xxx and STM32F43xxx and NA(RCC) – Updated <i>Section 6.1.1: System reset</i> and <i>Section 6.1.3: Backup domain reset</i>. – Updated ethernet PTP clock in <i>Figure 16: Clock tree</i>. – Added note regarding backup domain reset in BDRST bit of <i>RCC Backup domain control register (RCC_BDCR)</i>. – Added register reset values in <i>Table 38: RCC register map and reset values for STM32F42xxx and STM32F43xxx</i> Section 7: Reset and clock control for STM32F405xx/07xx and STM32F415xx/17xx(RCC) – Updated <i>Section 7.1.1: System reset</i> and <i>Section 7.1.3: Backup domain reset</i>. – Updated ethernet PTP clock in <i>Figure 21: Clock tree</i> – Added note regarding backup domain reset in BDRST bit of <i>RCC Backup domain control register (RCC_BDCR)</i>. – Added register reset values in <i>Table 39: RCC register map and reset values for STM32F405xx/07xx and STM32F415xx/17xx</i>

Table 315. Document revision history (continued)

Date	Version	Changes
05-Feb-2024	20 (continued)	Section 10: DMA controller (DMA) Updated DMA stream x FIFO control register (DMA_SxFCR) (x = 0..7) address offset. Section 12: Interrupts and events Changed Pending register (EXTI_PR) reset value to 0x0000 0000. Section 17: Advanced-control timers (TIM1 and TIM8) Updated Section 17.3.7: PWM input mode. Updated SMS in Section 17.4.3: TIM1 and TIM8 slave mode control register (TIMx_SMCR). Updated OC1PE in Section 17.4.7: TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1). Section 22: Window watchdog (WWDG) Updated t_{WWDG} equation in Section 22.4: How to program the watchdog timeout. Section 26: Real-time clock (RTC) – Updated HSE clock in Figure 237: RTC block diagram (NA devices). – Updated Section 26.3.6: Reading the calendar. Section 29: Serial audio interface (SAI) – In the whole section, replaced TDM by free protocol mode. – In Section 29.18.4: SAI x frame configuration register (SAI_XFRCR) where x is A or B, specified that FRL[7:0] must be configured when the audio block is disabled. Section 34: USB on-the-go full-speed (OTG_FS) – Updated Figure 392: Device-mode FIFO address mapping and AHB FIFO access mapping. – Modified OTG_FS USB configuration register (OTG_FS_GUSBCFG) OTG_FS_GUSBCFG reset value. Section 33: Ethernet (ETH): media access control (MAC) with DMA controller Updated bits that control the checksum in Section : Transmit checksum offload Section 38: Debug support (DBG) Removed note on APB bridge write buffer after Table 302: Flexible SWJ-DP pin assignment Updated REV_ID[15:0] in Section : DBGMCU_IDCODE Section 39: Device electronic signature Updated Section 39.1: Unique device ID register (96 bits) Added Section 40: Important security notice.

Table 315. Document revision history (continued)

Date	Version	Changes
07-Jun-2024	21	Updated the note in Bit[11:0] description of <i>Section 18.6.8: I²C Clock control register (I2C_CCR)</i> . Fixed typo in <i>Section 10.1.3: Interrupt and exception vectors</i> .
04-May-2026	22	Updated: – <i>Section 3.9.8: Flash control register (FLASH_CR) for STM32F42xxx and STM32F43xxx</i> – <i>Section 5.1.3: Voltage regulator for STM32F405xx/07xx and STM32F415xx/17xx</i> – <i>Section 13.13.1: ADC status register (ADC_SR)</i> – <i>Section 26.6.21: RTC register map</i> – Fixed typo in <i>Section 27: Inter-integrated circuit (I2C) interface</i> – OTG_FS Host non-periodic transmit FIFO size register (OTG_FS_HNPTXFSIZ)/Endpoint 0 Transmit FIFO size (OTG_FS_DIEPTXF0) in <i>Section 34: USB on-the-go full-speed (OTG_FS)</i>